

Lecture Notes: Formal/Game Theory

I. UTILITY

A. Ordinal Utility

Utility is a purposefully vague concept, but higher values denote greater gains from some outcome. For example:

$$X = \{a, b, s, w\}$$

$$b > s$$

$$s > w$$

$$w > a$$

bswa; yes, transitive



Preference order: b s w a

Can also have indifference:

$$b > s$$

$$s \mid w$$

$$w > a$$

bswa

bwsa

Evaluate transitivity; for example:

$$b > s$$

$$s > w$$

$$w > a$$

$$a > b$$

intransitive preferences

($b > s$, $s > w$, $w > a$, BUT $a > b$)

Multiple utility functions may capture the above ordering:

e.g., $U(a) = 0$, $U(b) = 10$, $U(s) = 5$, $U(w) = 2$

e.g., $U(a) = -100$, $U(b) = 1$, $U(s) = 0$, $U(w) = 0$

b s w a

Doesn't matter: can plug in either of the above for utility and preferences remain transitive.

B. Cardinal Utility

Values *do* matter once we introduce uncertainty. In cases where we only have probabilities of outcomes, we are now working with **expected utilities**.

For example, consider three candidates A, B, and C, with associated utilities:

$$U(A) = 1, U(B) = 0, U(C) = -2$$

So we'd vote for/donate to candidate A. Easy.

BUT, what if candidate A is a longshot? Let these be the probabilities of each winning, given our decision of whether to support A or B:

		Outcome		
		A Wins	B Wins	C Wins
Decision	Support A	0.1	0.4	0.5
	Support B	0	0.6	0.4

This introduces the idea of *expected* utility. To calculate expected utility, multiply the utility values by probabilities of outcomes:

$$E(U)_{\text{support A}} = (0.1 * 1) + (0.4 * 0) + (0.5 * -2) = -0.9$$

$$E(U)_{\text{support B}} = (0 * 1) + (0.6 * 0) + (0.4 * -2) = -0.8$$

II. GAMES

A. Prisoner's Dilemma:

		Player 2	
		C	D
Player 1	Cooperate C (don't talk)	1 year, 1 year	3 years, 0 years
	Defect D (rat out)	0 years, 3 years	2 years, 2 years

Nash equilibrium (NE)

$NE = \{D, D\}$

Informal

Information definition: given all other players' choices (holding them constant), each player's choice is rational/utility maximizing.

Formal definition: An action profile such that no player has a unilateral incentive to deviate.

NE does not explain how players will act, but only says that if reached, players have no incentive to leave.

Find NE for Prisoner's Dilemma game. Given column plays C, what is the best response for Player 1?

		Player 2		
		C	D	
Player 1	C	1 year, 1 year	3 years, 0 years	NE = {D,D} (where they intercept)
	D	0 years, 3 years	2 years, 2 years	

Political application: Tragedy of the Commons [whenever common resources, the problem of free-riding]

		Country 2	
		Cut	Don't Cut
Country 1	Cut	3,3	0,5
	Don't Cut	5,0	1,1

In-class extra credit example.

Elinor Olstrom's contribution: but people *do* manage common resources (fisheries, forests, waters, etc.) in many instances. Why?

Situational variables that engender trust and reciprocity (non-monetized relationships), or shame. One example is the social stigma attached to littering.

B. The Ultimatum Game

		Player 2	
		Accept	Deny
Player 1	Altruistic	50,50	0,0
	Selfish	99,1	0,0

Player 2 should always accept if offer > 0, but don't.

C. The minimax principle

Games can be designed in such a way to incentive fairness.

For example, the divide the cake game:

		Chooser	
		Choose Bigger	Choose Smaller
Cutter	Cut evenly	Half the cake - crumb, Half the cake + crumb	Half the cake - crumb, Half the cake + crumb
	Cut unevenly	Small piece, Big piece	Big piece, Small piece

Political application: John Rawls, the veil of ignorance and the difference principle.